Electrical build sequence and A1 changes

The A0 detailed procedure remains the reference for the common MCU, power converter, inputs, outputs, SWD, UART and USB circuits. Use A1 component placements and nets, not the A0 manufacturing files. First complete a routed, reviewed release and resolve purchasing placeholders.

For power-only bring-up, populate J1, F1, J17, D1, U2, C1–C3 and D3/R75. Wire and verify SW3 before applying the current-limited 24 V source. The original 100 mA power-only limit is a starting test setting, not a final board limit. Measure VIN24, VIN_FUSED, VIN_SWITCHED, V24 and +3V3. Switch off immediately if the source current-limits unexpectedly. Check PWR independently of MCU firmware; disconnect the source before adding parts. Do not jumper J17 in the final prototype: that bypasses the requested power switch.

After logic, SWD and I/O acceptance, add DS1 separately. Verify supply polarity, the effective I²C pull-ups and selected module address before enabling its driver. Implement the A1 status ownership and bounded-update policy in the design document; do not relabel the old Nucleo application LED as a working heartbeat. Check retained power from external I/O, UART, USB or debugger sources before treating the power switch as a convenient bench stop. It is not an emergency-stop or safety disconnect. Applications currently need re-download after reset; persistence remains future software work.